

GLS NEXUS HACKATHON 2026 • PROBLEM STATEMENT KDAC-4

CricAI Platform

ICC Men's T20 World Cup 2026 Match Outcome Prediction & Strategy Engine

Company: Kenex AI

Domain: Machine Learning & Data Analytics

Highlights: End-to-End Python Pipeline • Scikit-Learn ML Models

Why This Matters

The critical strategic necessity in modern T20 tournament analytics

Team Management & Operations

- Eliminates subjective human bias during playing XI selection and team composition choices.
- Monitors real-time player form trends and venue pitch suitabilities across tournament stages.
- Provides empirical backing for toss decisions based on atmospheric dew models.
- Replaces reactive post-match debriefs with predictive pre-match tactical simulations.

Broadcasters & Fan Engagement

- Delivers live match win probability shifts ball-by-ball during active broadcasts.
- Generates data-backed storylines for commentators, analysts, and pre-show panels.
- Provides transparent feature importance metrics explaining key win drivers.
- Enhances fan engagement with accessible probabilistic match breakdowns.

Data Analytics Infrastructure

- Automates 70%+ of manual data cleaning and transformation processing overhead.
- Unifies fragmented venue reports, weather telemetry, and scorecard event feeds.
- Provides instant access to clean feature matrices without manual SQL wrangling.
- Establishes a standardized data warehouse foundation for continuous model updates.

500+

Matches Analyzed

11,900+

Ball-by-Ball Records

17

Engineered Features

67%

Model Accuracy

Problem Statement

Fragmented data sources and lack of real-time strategic intelligence

The Data Fragmentation Challenge

Modern cricket generates massive volumes of detailed data, but it exists in isolated silos across multiple disparate formats:

- **Player Statistics:** Historical records for batting averages, strike rates, and bowling economies.
- **Venue & Weather Conditions:** Pitch pace, spin ratings, humidity, temperature, and dew factors.
- **Ball-by-Ball Feeds:** Detailed event logs capturing runs, boundaries, and wickets.
- **Squad Dynamics:** Team composition changes and role assignments across phases.

Consequence: Current match analysis is manual, partial, and performed too late during a tournament to influence real-time strategy.

Required Platform Deliverables

- **Data Source Simulation:** Generate T20 match feeds for 20 international teams across 10 venues using Python.
- **Data Warehouse Design:** Architect a clean SQLite relational database schema stitching player stats, matches, and weather.
- **Automated ETL Pipeline:** Automate data extraction, cleaning, transformation, and feature storage using Pandas.
- **ML Win Predictor:** Train Random Forest and Logistic Regression models for probabilistic match outcome predictions.
- **GenAI Strategy Layer:** Automatically synthesize model outputs into actionable tactical briefs for team management.
- **Streamlit Dashboard:** Interactive web interface for team analysis, live predictions, and strategy exports.

Proposed Solution: CricAI Ecosystem

An end-to-end platform bridging raw data to tactical execution

1. Data Engineering & Warehousing

- Simulates 500+ historical and live-style T20 matches across 20 international teams using Python.
- Processes 11,900+ granular ball-by-ball event records into structured formats.
- Cleans missing values, normalizes strike rates, and computes player performance indices using Pandas.
- Stores structured records in an operational SQLite database warehouse (`data/cricket\_warehouse.db`).

2. Predictive Machine Learning Engine

- Engineers 17 dynamic features spanning team ranking differentials, win rates, pitch ratings, and dew indices.
- Trains Random Forest and Logistic Regression classification models using Scikit-Learn with 5-fold cross-validation.
- Selects the optimal model based on accuracy and AUC-ROC evaluation metrics.
- Outputs real-time win probabilities %, model confidence scores, and key predictive drivers.

3. Strategy Brief & User Interface

- Translates quantitative win probability outputs into plain-language tactical coaching briefs.
- Provides venue-specific batting acceleration targets, bowling containment plans, and toss guidance.
- Delivers interactive EDA analytics, scenario simulation, and 1-click strategy exports via Streamlit.
- Exposes real-time prediction endpoints through a lightweight REST API built with FastAPI.

Unique Selling Proposition (USP)

Competitive analysis demonstrating why CricAI stands out

Capability / Feature	CricAI Platform	ESPNcricinfo / Public Tools	Traditional Scouting
End-to-End Automated Pipeline	✓ Fully Automated Python ETL & ML	✗ Static / Manual Updates	✗ Spreadsheets
Environmental Physics Integration	✓ Dew, Pitch Pace/Spin, Wind, Temp	✗ Basic Pitch Type Only	✗ Subjective Reports
GenAI Tactical Strategy Briefs	✓ Automated Coaching Briefs	✗ None	✗ Manual Notes
Interactive Dashboard UI	✓ Streamlit Web Application	✗ Closed Proprietary UI	✗ None
Explainable Key Factors	✓ Feature Importance Breakdown	✗ Black-box Odds	✗ Intuition-based

Core Competitive Advantage: CricAI is an open, easy-to-deploy platform combining quantitative win probabilities with natural language AI coaching notes using a clean, accessible Python tech stack.

System Architecture & Workflow

Data transformation journey from ingestion to strategic presentation



Phase 1: Data Ingestion & Storage

Simulated match feeds and ball-by-ball logs are ingested, cleaned using Pandas, and stored in a lightweight SQLite relational database.

Phase 2: ML Model Training

17 engineered differential features feed a Scikit-Learn Random Forest model to predict win probabilities with 5-fold cross-validation.

Phase 3: Streamlit UI & Strategy

Model outputs are presented through an interactive Streamlit dashboard featuring live predictions and plain-language strategy briefs.

Simplified Technology Stack

Clean, robust Python tools for easy development and deployment

Data Processing

- **Python 3.11:** Main programming language.
- **Pandas:** Data cleaning, transformation, and aggregation.
- **NumPy:** Fast numerical operations and synthetic statistical data simulation.

Data Storage

- **SQLite Database:** Lightweight, serverless relational database (``cricket_warehouse.db``).
- **6 Relational Tables:** ``teams``, ``players``, ``matches``, ``venue_conditions``, ``innings_summary``, ``ml_features``.

Machine Learning

- **Scikit-Learn:** Random Forest and Logistic Regression ML classification models.
- **Cross-Validation:** Stratified 5-fold validation for evaluation.
- **Feature Importances:** Driver extraction for explainability.

User Interface & API

- **Streamlit:** Interactive web dashboard UI.
- **Plotly:** Interactive charts and graphs.
- **FastAPI:** Lightweight REST backend.
- **Git & GitHub:** Version control repository.

SQLite Data Warehouse Architecture

Simple normalized relational database schema supporting analytical queries

teams Table

- team_id (Primary Key, INTEGER)
- team_name (TEXT)
- icc_ranking (INTEGER)
- win_rate_last_2_years (REAL)
- avg_team_score_t20 (INTEGER)
- nrr (Net Run Rate, REAL)
- home_win_advantage (REAL)

Stores core team performance metrics, rankings, and historical winning percentages.

matches Table

- match_id (Primary Key, INTEGER)
- date (TEXT)
- team_a_id, team_b_id (Foreign Keys)
- venue_name, pitch_type (TEXT)
- toss_winner_id, toss_decision (TEXT)
- score_team_a, score_team_b (INTEGER)
- winner_id (Foreign Key, INTEGER)

Stores individual match scorecards, venue linkages, toss decisions, and match outcomes.

venue_conditions Table

- match_id (Foreign Key, INTEGER)
- temperature_c (REAL)
- humidity_pct (REAL)
- wind_speed_kmh (REAL)
- dew_factor_numeric (INTEGER 0-3)
- pitch_pace_rating (REAL)
- pitch_spin_rating (REAL)

Captures match-level weather telemetries and pitch ratings used as inputs for ML feature engineering.

Machine Learning Predictive Engine

Feature engineering, model training, and accuracy benchmarks

Predictive Pipeline Steps

- **1. Data Preparation:** Extract 17 match metrics including ranking differentials, 2-year win rate delta, pitch spin ratings, and dew factors using Pandas.
- **2. Encoding & Cleaning:** Encode categorical pitch and toss attributes using Scikit-Learn LabelEncoder; fill missing data using median values.
- **3. Cross-Validation:** Evaluate models using Stratified 5-Fold Cross-Validation to ensure unbiased performance.
- **4. Model Fitting:** Fit Random Forest (200 decision trees) and Logistic Regression classifiers.
- **5. Model Output:** Generate win probability percentages %, model confidence scores, and top predictive drivers.

Model Performance Evaluation

Model Name	Test Accuracy	AUC-ROC
Random Forest (Selected Best)	67.0%	0.669
Logistic Regression	66.0%	0.665

Key Predictive Factors: Recent win rate differential (7.8%), wind speed (7.7%), pitch spin rating (7.5%), temperature (7.4%), pitch pace rating (7.2%), and NRR differential (7.0%).

GenAI Tactical Strategy Generator

Automated natural language coaching briefs for match preparation

Strategy Generation Logic

- Receives probabilistic win outputs and confidence scores directly from the Scikit-Learn ML engine.
- Analyzes match parameters including pitch spin/pace scores, humidity levels, and dew factor indicators.
- Synthesizes pitch-specific batting acceleration targets, bowling containment tactics, and toss recommendations.
- Generates automated, structured briefing notes for head coaches, team captains, and performance analysts.
- Built using clean Python text generation logic with optional LLM API compatibility.

Sample Strategy Brief Output

```
≡ STRATEGY BRIEF - India vs Pakistan ≡
Phase: Semi-Final | Pitch: Spin-Friendly | Dew: Medium

🏏 PREDICTION: India (68.3% Win Probability | Confidence: 36.5/100)

🔧 BATTING STRATEGY (India):
• Promote aggressive lower-order hitter for final 4 overs.
• Use spin-heavy middle overs to build steady total.

🎯 BOWLING STRATEGY (India containment plan for Pakistan):
• Deploy premier spinner from over 3 – surface will grip early.
• Target top-order batsmen with flight and turn.

📊 TOSS INSIGHT:
• Fielding first is strategically sound – dew aids chasing team in 2nd
innings.
```

Interactive Streamlit Dashboard

Visual exploratory data analysis and live prediction GUI

1. Team & Venue Analytics

Interactive visual analytics displaying team win rates, head-to-head records, venue scoring averages, pitch distributions, and toss decision charts built with Plotly.

- Team ranking vs average score scatter plots.
- Venue scoring history horizontal bar charts.

2. Match Predictor GUI

Interactive web interface allowing users to select playing teams, select tournament phases, adjust pitch pace/spin ratings, and simulate dew conditions in real time.

- Real-time win probability bar visualizations.
- Model confidence gauge score display.

3. Strategy Brief Export

Automated strategy generation panel displaying plain-language coaching notes with 1-click text file download functionality for pre-match briefing sessions.

- Instant strategy brief generation.
- 1-click text export for team staff.

Deployment & Execution

Easy local execution and 1-click cloud deployment

Simple Local Execution (3 Commands)

- **Step 1: Install Dependencies**

```
pip install -r requirements.txt
```

- **Step 2: Run Automated Pipeline**

```
python run_demo.py (simulates data, builds database, trains ML model)
```

- **Step 3: Launch Streamlit Web App**

```
streamlit run src/dashboard/app.py
```

1-Click Cloud Deployment Options

- **Streamlit Community Cloud (Recommended):** Connect GitHub repository and deploy `src/dashboard/app.py` instantly for free.
- **Render / Railway:** Deploy as a web service using the included Dockerfile.
- **FastAPI REST Backend (Optional):** Run `uvicorn src.api.main:app --reload` to access API endpoints and Swagger docs.

Assumptions, Challenges & Future Scope

Evaluation considerations and technical roadmap

Key Project Assumptions

- Synthetic dataset parameters accurately mirror international T20 match distributions.
- Teams field optimal playing XIs without last-minute unannounced player injuries.
- Venue pitch characteristics follow standard historical regional profiles.

Technical & Operational Challenges

- Quantifying sudden micro-weather shifts and rapid humidity changes mid-innings.
- Securing free, open real-time live ball-by-ball streaming API feeds during broadcasts.
- Balancing feature weights between long-term historical records and recent form.

Strategic Future Scope

- Direct integration with Cricsheet API for automatic real-world match data updates.
- Computer Vision pitch surface inspection for automated pace/spin rating calculation.
- Fine-tuned LLM for live ball-by-ball strategic commentary generation.

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Thank You!

CricAI — Data Analytics & AI Strategy Engine for T20 Cricket

Company: Kenex
AI

Platform: CricAI End-to-End Prediction
Platform

Deliverables: Full Codebase • Data Warehouse • Scikit-Learn Model • Streamlit
App